Supplementary Information

Stage-Resolved Transcriptomic Analysis Reveals Distinct Molecular Programs Associated with Liver Fibrosis Severity

Authors: [AUTHOR NAMES AND AFFILIATIONS]

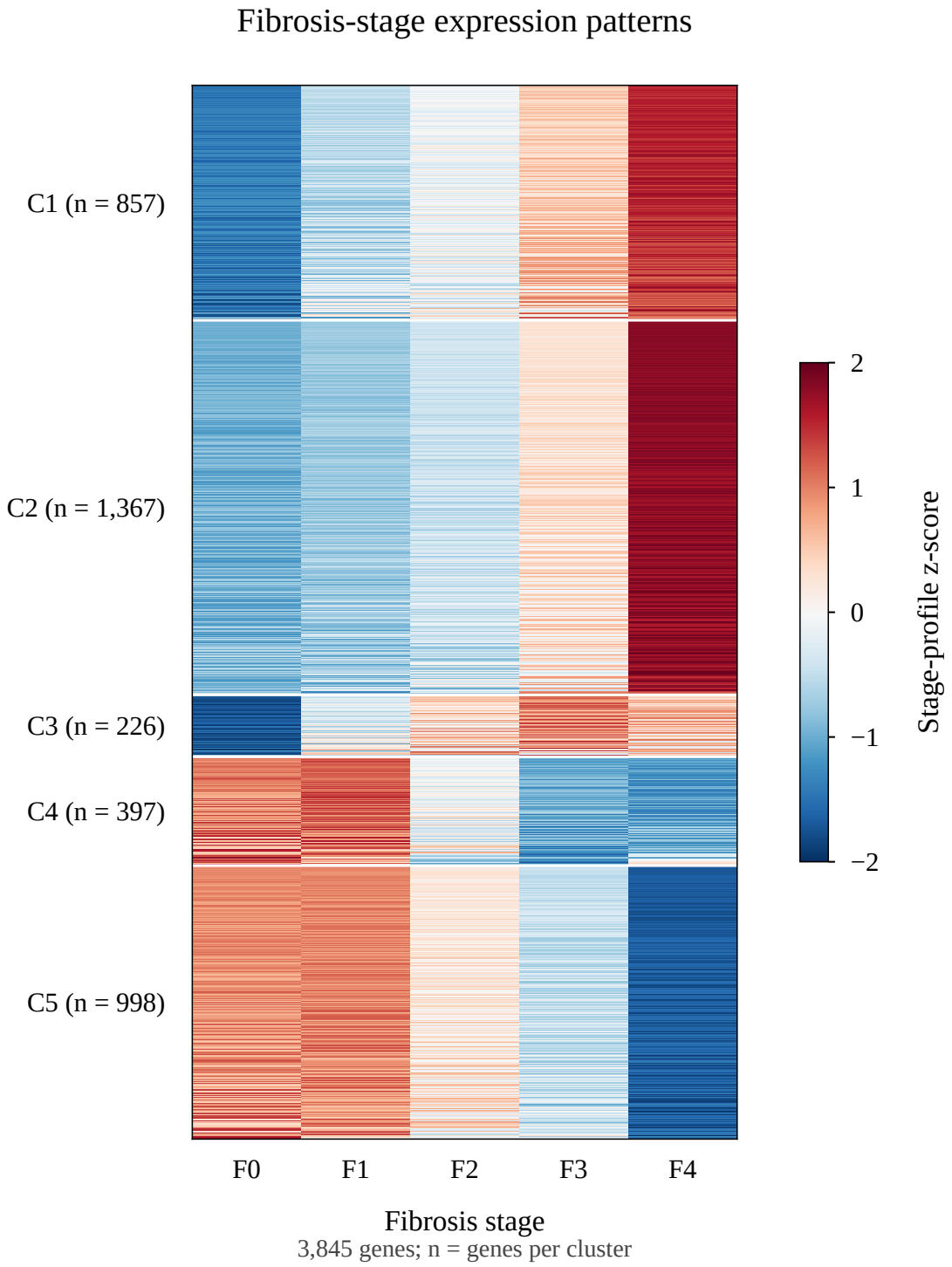


Figure S1: Fibrosis-stage expression-profile heatmap. Row-standardised profiles of 3,845 fibrosis-associated genes across F0–F4, grouped into five clusters. The colour scale represents relative expression across stages; the profiles describe cross-sectional differences between patients.

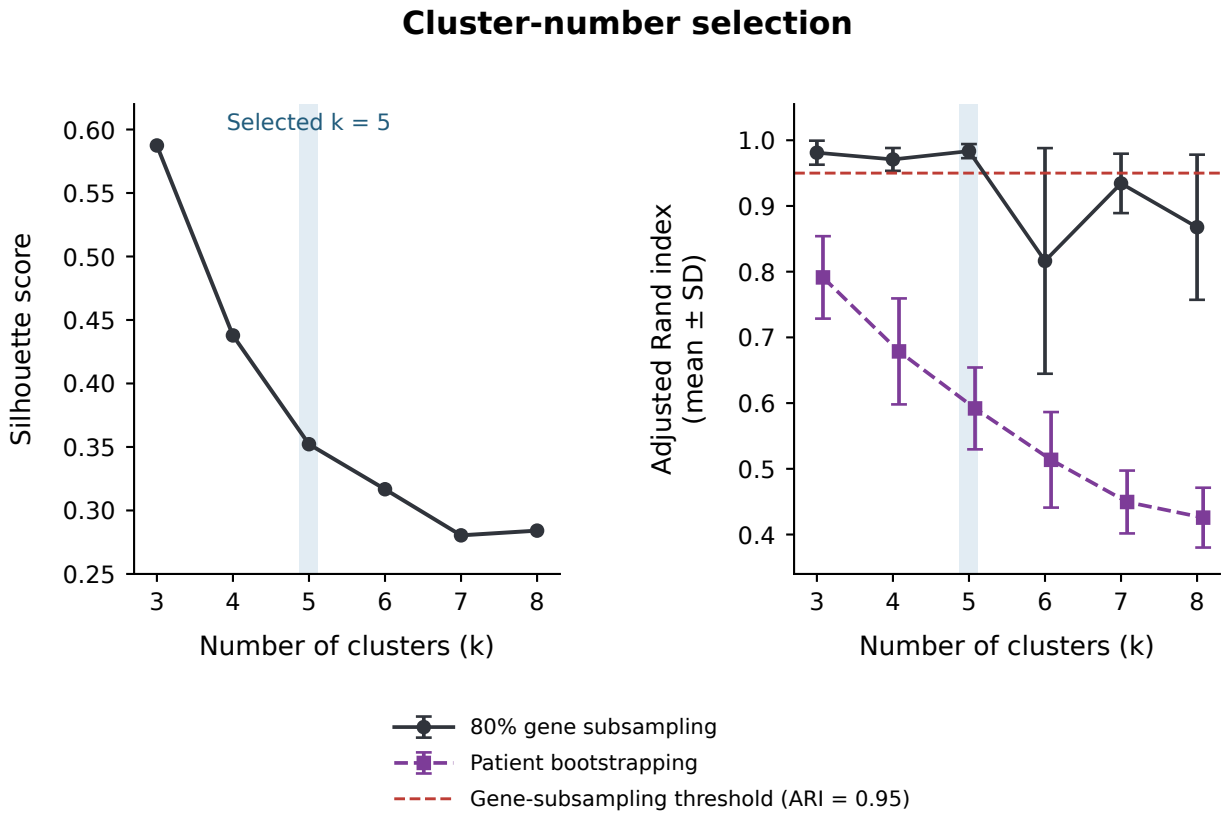


Figure S2: Cluster-number diagnostics. Silhouette scores and adjusted Rand indices for $k = 3$–8. Stability error bars show SD across 50 gene-subset or patient-bootstrap refits. The selected $k = 5$ was the largest solution with mean gene-subset ARI ≥ 0.95; patient-bootstrap stability was assessed separately. Numerical values are reported in Supplementary Table S1.

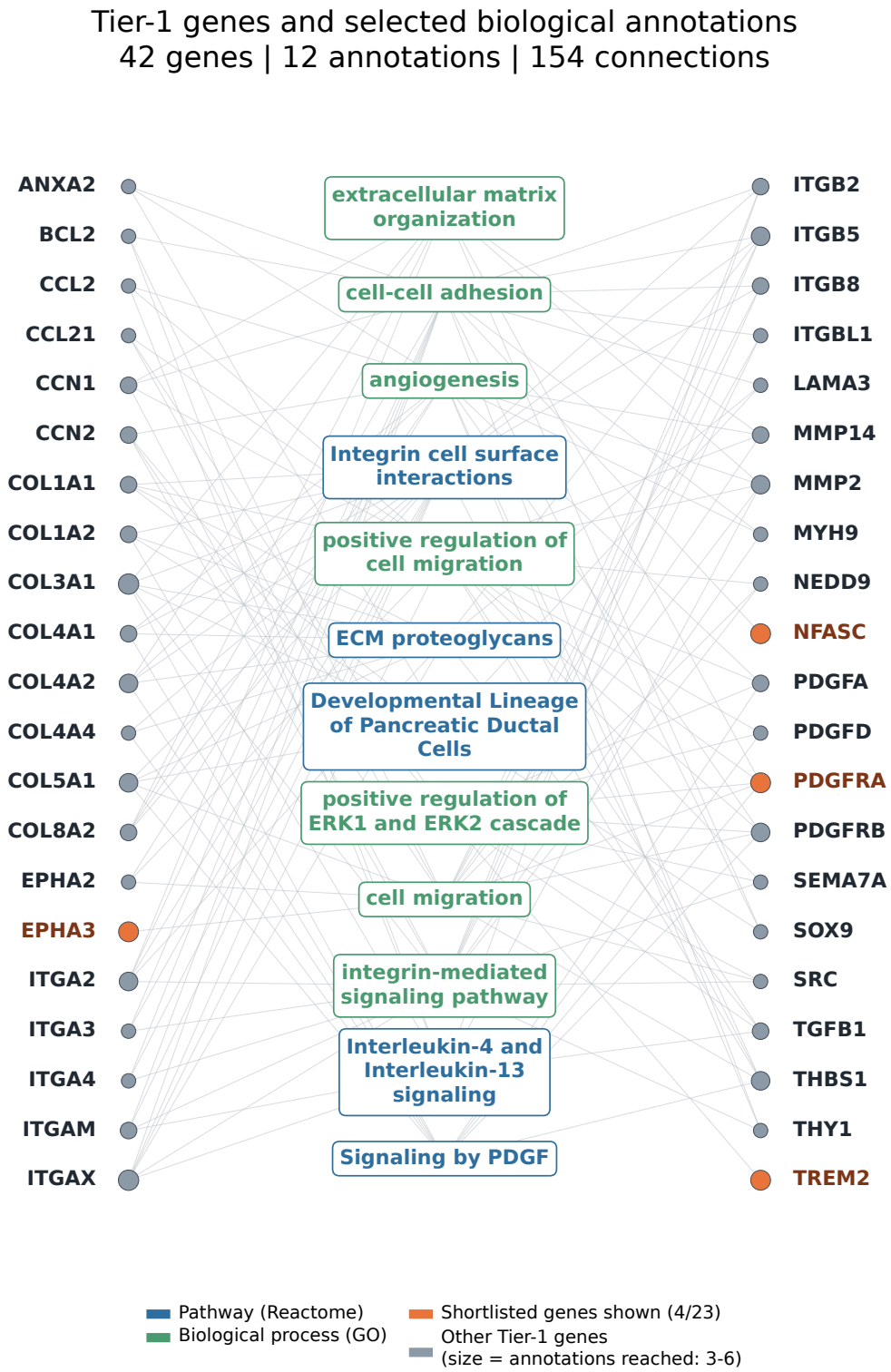


Figure S3: Prioritised gene–annotation network. Curated connections link 42 Tier 1 genes to twelve annotations through 154 relationships. Displayed genes reach at least three selected annotations, or are shortlisted genes reaching at least one. Connections represent annotation membership, not demonstrated physical or causal interactions.

Table S1: Cluster-number diagnostics for all evaluated solutions

k	Silhouette	Gene-subset ARI (mean ± SD)	Patient-bootstrap ARI (mean ± SD)	Gene-subset mean ARI ≥ 0.95
3	0.5874	0.9811 ± 0.0183	0.7913 ± 0.0627	Yes
4	0.4378	0.9708 ± 0.0174	0.6787 ± 0.0806	Yes
5	0.3522	0.9835 ± 0.0107	0.5919 ± 0.0623	Yes
6	0.3167	0.8164 ± 0.1717	0.5137 ± 0.0727	No
7	0.2804	0.9343 ± 0.0452	0.4495 ± 0.0478	No
8	0.2841	0.8676 ± 0.1104	0.4257 ± 0.0455	No

Gene-subset diagnostics used 50 refits with 80% of genes; patient-bootstrap diagnostics used a separate 50 refits resampling patients within studies. Values are reproduced from the cluster-number diagnostics, without reclustering. The selection rule chose the largest k with mean gene-subset ARI at least 0.95: k = 5. The highest silhouette score occurred at k = 3; patient-bootstrap ARI was not the selection criterion. Means and SDs are rounded to four decimal places for display; eligibility was assessed using the unrounded means.

Table S2: Dataset-specific preprocessing and final analysis eligibility

Dataset	Expression processing	Sample handling and eligibility	Fibrosis n	Disease group n
GSE126848	Raw counts; CPM rescaling; $\log_2(\text{CPM}+1)$	Twelve obese-only samples excluded from ordered disease-group analysis. Fibrosis stages unavailable.	0	45
GSE130970	TPM; $\log_2(\text{TPM}+1)$	Five ambiguous disease classifications marked unclassified; fibrosis stages retained. Remaining disease labels are histology-derived proxies.	78	73
GSE135251	Deposited counts; CPM rescaling; $\log_2(\text{CPM}+1)$	Fibrosis annotations retained. Broad NAFLD labels excluded from ordered disease-group analysis, leaving only controls and no variation on that axis.	216	0
GSE162694	Raw counts; CPM rescaling; $\log_2(\text{CPM}+1)$	Fibrosis annotations retained. Broad NAFLD labels excluded from ordered disease-group analysis, leaving only controls and no variation on that axis.	143	0
GSE167523	Raw counts; CPM rescaling; $\log_2(\text{CPM}+1)$	All 98 NAFL/NASH-labelled observations eligible. Fibrosis stages unavailable.	0	98
GSE193066	Deposited RLE-normalised counts; $\log_2(\text{count}+1)$; no additional CPM rescaling	164 biopsies resolved to 106 patient observations: 28 same-stage pairs averaged; 30 different-stage pairs represented by the first biopsy. No usable disease-group labels.	106	0
GSE240729	Raw counts; CPM rescaling; $\log_2(\text{CPM}+1)$	All fibrosis-staged observations retained. No usable disease-group labels.	67	0
GSE269412	Raw counts; CPM rescaling; $\log_2(\text{CPM}+1)$	Included in the expression resource; required clinical annotations unavailable for either association analysis.	0	0
Total	—	Final analytical observations within studies	**610**	**216**

Counts refer to patient-level analytical observations within each study. Participant independence across accessions was not established. The two analysis populations overlap and must not be added together. Zero indicates no contribution to the specified association analysis, rather than exclusion from the expression resource. Input sample counts and source filenames are listed in Table 3.

For all datasets, duplicate raw gene identifiers were resolved by retaining the first occurrence. Gene identifiers were harmonised to Ensembl identifiers, and measurements from distinct identifiers mapping to the same sample–gene pair were summed before transformation. CPM library totals were calculated before mapping and filtering. Transformed expression was standardised per gene within each study (ddof $= 1$), followed by patient-level biopsy selection or averaging. GSE193066 standardisation used all 164 source biopsies before repeat-biopsy resolution. Study eligibility required at least 25 patient observations and variation on the relevant clinical axis.

Table S3: Gene-prioritisation component scores and decision rules

Score	Evidence	Assignment rule
P_g	Fibrosis-association magnitude	**2**: $\lvert$ median $\rho\rvert \geq 0.40$. **1**: $0.30 \leq \lvert$ median $\rho\rvert < 0.40$. **0**: otherwise. Applied to genes already passing fibrosis-association discovery criteria.
T_g	F0–F4 endpoint separation	Let $\Delta_g = \overline{z}_{F4} - \overline{z}_{F0}$. First require direction agreement with the median fibrosis correlation: both values positive or both non-positive. If agreement holds, assign **2** for $\lvert\Delta_g\rvert \geq 1$, **1** for $0.5 \leq \lvert\Delta_g\rvert < 1$, and **0** otherwise. Direction disagreement assigns **0**. A zero difference receives **0** by magnitude. Cluster direction does not override this rule.
X_g	Largest adjacent-stage contrast and matched-refit reproducibility	Select the full-data comparison with the largest available $\lvert\beta\rvert$. Define A: $q < 0.05$ and $\lvert\beta\rvert \geq 0.20$ in that comparison. Define B: at least 50% of matched draws have $\lvert\beta\rvert \geq 0.20$ and the same sign as the selected full-data coefficient. Assign **2** if A and B hold, **1** if exactly one holds, and **0** if neither holds. Matched-draw significance does not contribute.
F_g	Operational fibrosis-only membership	**1**: selected in the fibrosis analysis but not in the ordered disease-group analysis. **0**: otherwise. This selection status does not establish biological specificity.
B_g	Functional-annotation support	Count annotation collections containing at least one significantly enriched term (BH $q < 0.05$) whose overlapping-gene list contains the gene, using its own concordant direction group or its full expression cluster. **2**: at least two collections; **1**: one; **0**: none. Eligible collections: GO Biological Process, KEGG legacy, Reactome and Hallmark. Each counts once across both routes; confidence-core enrichment is excluded.

Precision and thresholds. Scoring uses exported median correlations and stage means recorded to three decimal places; the endpoint difference is also rounded to three decimals before comparison. Lower score thresholds are inclusive; statistical significance requires $q < 0.05$.

Contrast selection and missing evidence. Exact ties in $\lvert\beta\rvert$ retain the first comparison in the order F0–F1, F1–F2, F2–F3, F3–F4. A missing full-data q-value does not satisfy condition A. Missing matched evidence for the selected contrast, or absence of a selected contrast, stops scoring rather than automatically assigning zero. All 3,845 fibrosis-associated genes had finite full-data coefficients and q-values for all four contrasts and matched evidence for the selected contrast.

Annotation eligibility. All significant terms are eligible, including terms not displayed in figures. Discordant shared genes have no concordant direction-group route but can receive support through their full expression cluster. Multiple terms within one collection do not increase its contribution.

Total score and shortlist. $S_g = P_g + T_g + X_g + F_g + B_g$ has a maximum of nine. Tier 1: $S_g \geq 7$; Tier 2: $5 \leq S_g < 7$; Tier 3: $S_g < 5$. Up to five Tier 1 genes per cluster are selected by descending total score, then descending absolute median fibrosis correlation. Clusters with fewer than five qualifying genes contribute fewer candidates. The components share evidence and do not constitute independent validation.

Table S4: Final integrated graph composition and drug-annotation coverage

Measure	Count
Total nodes	126,763
Total relationships	35,138,047
Broad entity categories	8
Relationship types	14
Drug nodes	12,544
OptimusKG-derived Drug subset	12,025
Drug nodes from ChEMBL/DGIdb/GtoPdb/Open Targets integration	519
ACTS_ON relationships (all sources)	25,890
OptimusKG-derived drug–gene records	20,674
Drug–gene records from ChEMBL/DGIdb/GtoPdb/Open Targets integration	5,216
Common source-list field	25,890
Common action-list field	25,430
Links from the four-database integration involving the 519 drugs	846
Links from the four-database integration involving OptimusKG drugs	4,370 (2,104 drugs)
Expression relationships	32,471,042
Fibrosis / disease-group progression links	3,845 / 4,199
Displayed gene–annotation subgraph	42 genes; 12 annotations; 154 links
Populated fields among all 12,544 Drug nodes	
SMILES	9,036
Molecule type	12,051
Maximum clinical phase	10,293
Approval flag	9,902
Withdrawal flag	11,623
Recovered name	10,878
Populated fields on the 20,674 OptimusKG-derived drug–gene records	
Direct-source, assertion, conflict and evidence-level fields	20,674 each
Reference URLs	2,893
Mechanism text	2,898

Counts were verified by direct queries of the integrated graph on 28 October 2026, including 12,544 Drug nodes and 25,890 ACTS_ON relationships. Progression-link endpoints and reported properties were verified against the analysis exports for all 8,044 links; the displayed subgraph was verified by gene identifier, annotation identifier and relationship type. These checks establish consistency with the reported analyses rather than independent biological validation.

Drug-source provenance. Drug–gene evidence records GtoPdb 2026.3.1; DGIdb 5, interactions downloaded in October 2026; ChEMBL REST API, October 2026; and Open Targets Platform 26.09. These identifiers are reproduced from graph properties. The metadata-recovery implementation distinguishes OptimusKG recovery from ChEMBL gap filling.

Coverage interpretation. A populated approval or withdrawal flag may be true or false. A phase field does not establish current approval, availability or efficacy for liver disease. All drug entities use the Drug label; source properties distinguish their provenance. Missing metadata remain missing; property coverage uses the stated denominator. Drug annotations were not inputs to the transcriptomic association, clustering or P/T/X/F/B prioritization procedures.

Supplementary Methods: implementation checks and reference-file inventory

Ambiguous disease classifications. The five GSE130970 samples assigned unclassified disease-group status in the primary analysis were GSM3758028, GSM3758038, GSM3758039, GSM3758073 and GSM3758074. Their original histological fields and fibrosis stages were retained.

Relationship schema. The integrated graph used 14 relationship types: EXPRESSES, IN_DATASET, ASSOCIATED_WITH, INTERACTS_WITH, INVOLVED_IN, IN_PATHWAY, ACTS_ON, TREATS, CONTRAINDICATED_IN, OFF_LABEL_FOR, HAS_PHENOTYPE, PARENT_OF, TRACKS_FIBROSIS and TRACKS_INFLAMMATION.

For transition scoring, the comparison with the largest absolute available full-data coefficient was selected, with ties resolved in the order F0–F1, F1–F2, F2–F3, F3–F4. A missing full-data q-value did not satisfy the substantial-difference condition. Missing matched evidence for the selected comparison, or the absence of a selected comparison for a gene, stopped scoring through an assertion rather than being assigned zero. In the final analysis, all 3,845 fibrosis-associated genes had finite coefficients and q-values for all four comparisons and matched evidence for the selected comparison. The reconstructed P/T/X/F/B scores agreed with the analysis table for all 3,845 genes.

The reference files read by the analysis scripts were data/raw/hgnc_complete_set.txt (file modification date 20 August 2026; most recent gene-record modification date within the file 18 August 2026) and the OptimusKG parquet files in data/optimuskg (file modification date 1 September 2026). SHA-256 checksums for all files are provided in the source manifest, source_checksums.txt. File dates and checksums identify these retained files and do not establish official release dates.